

## Performance Testing

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Date</b>          | 15 July 2026                            |
| <b>Team ID</b>       | XXXXXX                                  |
| <b>Project Name</b>  | Human Development Index (HDI) Predictor |
| <b>Maximum Marks</b> | 5 Marks                                 |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                   |
|----------------------------|---------------------------|
| <b>Testing Tool Used</b>   | Postman                   |
| <b>Type of Testing</b>     | Load Testing              |
| <b>Target Module / API</b> | HDI Prediction Module (/) |
| <b>Test Environment</b>    | Local (Flask Server)      |
| <b>Test Date</b>           | 15 July 2026              |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description    | No. of Virtual Users | Duration (sec) | Expected Outcome           |
|------|--------------------------------|----------------------|----------------|----------------------------|
| 1    | Predict HDI with valid inputs  | 1                    | 10             | HDI predicted successfully |
| 2    | Multiple prediction requests   | 10                   | 30             | Stable response            |
| 3    | Invalid input testing          | 5                    | 20             | Input validation           |
| 4    | Continuous prediction requests | 20                   | 60             | No application crash       |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks       |
|------|----------------------|--------------|--------------|----------------------|---------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 0.42 sec     | Pass                 | Fast response |
| 2    | Response Time (Max)  | < 5 seconds  | 1.12 sec     | Pass                 | Within limit  |
| 3    | Throughput (Req/sec) |              | 18 Req/sec   | Pass                 | Stable        |
| 4    | Error Rate           | < 1%         | 0%           | Pass                 | No errors     |
| 5    | CPU Utilization      | < 80%        | 28%          | Pass                 | Low usage     |
| 6    | Memory Utilization   | < 80%        | 36%          | Pass                 | Normal        |

### Step 4: Observations & Analysis

#### Key Findings

The HDI Predictor responded quickly under normal load and successfully generated predictions for all valid requests.

#### Bottlenecks Identified

No major bottlenecks were observed during local testing.

#### Optimization Steps Taken

- Cleaned dataset before training.
- Used a pre-trained Joblib model.
- Optimized Flask prediction logic for faster response.

### Step 5: Screenshots / Evidence

#### Human Development Index Predictor






**Predicted HDI : 1.0**

**Very High Human Development**

```
PS C:\Users\dimpl\Downloads\HDI-Predictor\HDI-Predictor> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 492-382-035
```